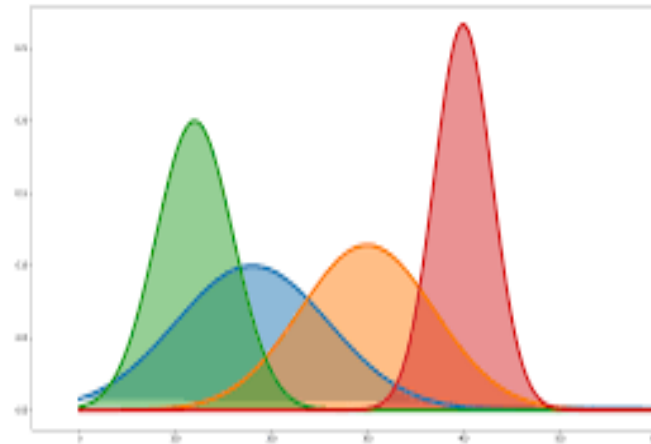


Week 5 Seminar Solution

Data Sampling and Confidence Intervals



Recall from lectures...

From Description to Inference: We will assume we have a bit of the data, but that we're missing most of it.

A bit of terminology:

- The **population** has **parameters**: μ is the mean; σ is the standard deviation; π is a proportion.
- Our **sample** has **statistics**.

<i>Parameter</i>	<i>Population</i>	<i>Sample</i>
Mean	μ	$\bar{x}$
Standard deviation	σ	s
Proportion	π	p

Recall from lectures...

Example:

We are asked to find the mean height of all first-year business and management students at the University of York (population).

- We measure all students in our class (the sample)
- We take the mean height of this sample
- Is this class a representative sample? If yes, the sample mean is an unbiased estimator of the population mean.

Recall from lectures...

Example:

We are asked to find the mean height of all first-year business and management students at the University of York (population).

- We measure all students in our class (the sample)
- We take the mean height of this sample
- Is this class a representative sample? If yes, the sample mean is an unbiased estimator of the population mean.
- Sampling: pros and cons, we need to be very careful.
- **Primary vs Secondary:** Primary data is any you collect yourself for this purpose. Secondary data is any data someone else has already collected.
- **Types of sampling:** Random, Systematic, Stratified, Cluster
- **Data collection:** closed-ended vs open-ended questions

Recall from lectures...

The Central Limit Theorem: No matter what the actual population distribution is, if you take many, large enough, samples and calculate the mean of each of those samples, those means will follow a normal distribution. This normal distribution has as its mean, the real population mean μ , and as its standard deviation, $\sigma/\sqrt{n}$, called the **standard error**.

❖ **Remember!** The **standard deviation** tells us about the **population** spread. The **standard error** tells us how spread out the means of **samples** are.

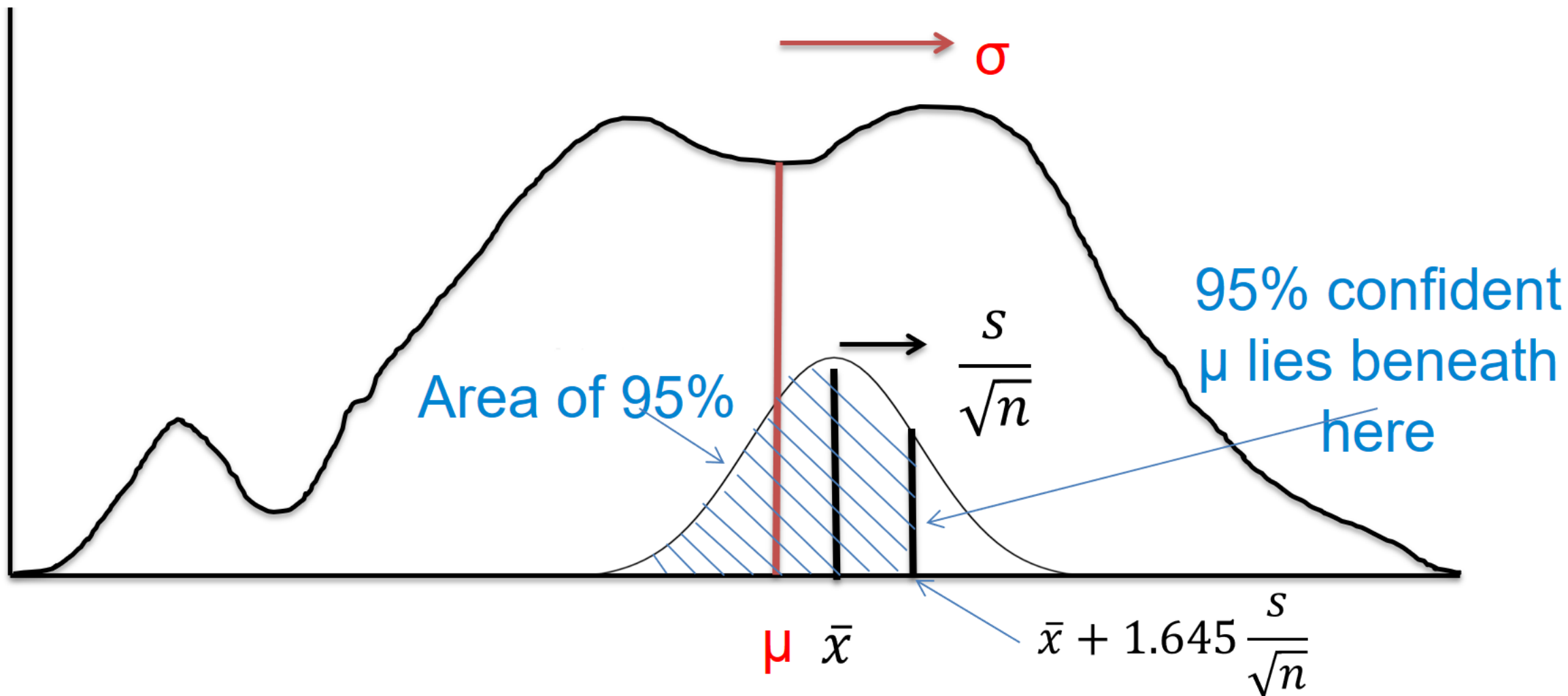
Recall from lectures...

The Central Limit Theorem: No matter what the actual population distribution is, if you take many, large enough, samples and calculate the mean of each of those samples, those means will follow a normal distribution. This normal distribution has as its mean, the real population mean μ , and as its standard deviation, $\sigma/\sqrt{n}$, called the **standard error**.

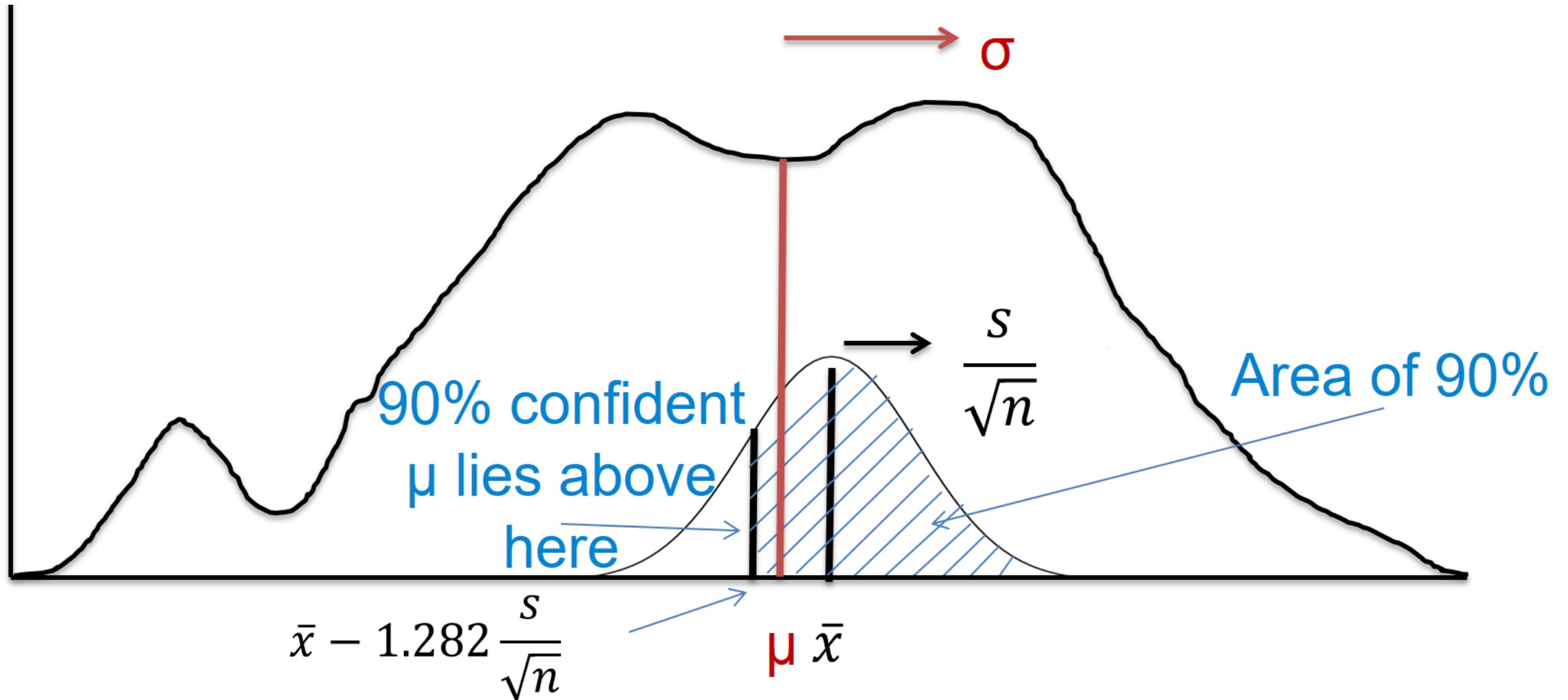
❖ **Remember!** The **standard deviation** tells us about the **population** spread. The **standard error** tells us how spread out the means of **samples** are.

- The CLT allows us to create prediction intervals for means. We call them “confidence intervals”. These interval estimates give us a good sense of accuracy.
- For example: We have 90% confidence that the population mean is between 59.6 and 60.4 (i.e. 60 ± 0.4).
- How does this help? We’re using $\bar{x}$ and s to estimate where μ is.

Upper bound example



Lower bound example



Recall from lectures...

Two issues: The **CLT** said “large enough” sample.

- What’s large enough? we tend to go with above 30... ish
- What if the sample isn’t “large enough”? we use the **t-distribution**

Recall from lectures...

Two issues: The **CLT** said “large enough” sample.

- What’s large enough? we tend to go with above 30... ish
- What if the sample isn’t “large enough”? we use the **t-distribution**
- **We are ready to construct a confidence interval!**

Format: Point Estimate $\pm$ Confidence Coefficient $\times$ Standard Error

$$\text{or } \bar{x} \pm CC \times \frac{s}{\sqrt{n}} \quad (\text{and } p \pm CC \times \sqrt{\frac{p(1-p)}{n}} \text{ if we are in a Bernoulli distribution})$$

- For example, for an interval of 95% with a sample size of 10, the Confidence Coefficient is:

Recall from lectures...

■ For example, for an interval of 95% with a sample size of 10, the Confidence Coefficient is:

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail Two Tails	5% 10%	2.5% 5%	1% 2%	0.5% 1%	0.1% 0.2%	0.05% 0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Recall from lectures...

■ For example, for an interval of 95% with a sample size of 10, the Confidence Coefficient is:

- ✓ for interval: two tails (for one of the bounds only, one tail)
- ✓ 5%
- ✓ Degrees of freedom: $n - 1 = 9$

■ For a sample of 100, wanting 99% confidence and just a lower bound, the CC number is:

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail Two Tails	5% 10%	2.5% 5%	1% 2%	0.5% 1%	0.1% 0.2%	0.05% 0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Recall from lectures...

■ For example, for an interval of 95% with a sample size of 10, the Confidence Coefficient is:

- ✓ for interval: two tails (for one of the bounds only, one tail)
- ✓ 5%
- ✓ Degrees of freedom: $n - 1 = 9$
- ✓ 2.262

■ For a sample of 100, wanting 99% confidence and just a lower bound, the CC number is:

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail Two Tails	5% 10%	2.5% 5%	1% 2%	0.5% 1%	0.1% 0.2%	0.05% 0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Recall from lectures...

- For example, for an interval of 95% with a sample size of 10, the Confidence Coefficient is: 2.262 (two tails, 5%, $n-1 = 9$)
- For a sample of 100, wanting 99% confidence and just a lower bound, the CC number is: 2.326 (one tail, 1%, $n-1 = 99 > 60$)

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail Two Tails	5% 10%	2.5% 5%	1% 2%	0.5% 1%	0.1% 0.2%	0.05% 0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Recall from lectures...

- For example, for an interval of 95% with a sample size of 10, the Confidence Coefficient is: 2.262 (two tails, 5%, n-1 = 9)
- For a sample of 100, wanting 99% confidence and just a lower bound, the CC number is: 2.326 (one tail, 1%, n-1 = 99 > 60)

Example:

A chemical process yields, over a sample of runs, products with masses: 2.3mg, 3.9mg, 1.9mg, and 2.7mg.
Calculate the 95% confidence interval for the true mean mass of the yield.

Remember: $\bar{x} \pm CC \times \frac{s}{\sqrt{n}}$

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail Two Tails	5% 10%	2.5% 5%	1% 2%	0.5% 1%	0.1% 0.2%	0.05% 0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Recall from lectures...

- For example, for an interval of 95% with a sample size of 10, the Confidence Coefficient is: 2.262 (two tails, 5%, n-1 = 9)
- For a sample of 100, wanting 99% confidence and just a lower bound, the CC number is: 2.326 (one tail, 1%, n-1 = 99 > 60)

Example:

A chemical process yields, over a sample of runs, products with masses: 2.3mg, 3.9mg, 1.9mg, and 2.7mg.
Calculate the 95% confidence interval for the true mean mass of the yield.

Remember: $\bar{x} \pm CC \times \frac{s}{\sqrt{n}}$

1st step: calculate the mean and the standard deviation:

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail Two Tails	5% 10%	2.5% 5%	1% 2%	0.5% 1%	0.1% 0.2%	0.05% 0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Recall from lectures...

- For example, for an interval of 95% with a sample size of 10, the Confidence Coefficient is: 2.262 (two tails, 5%, n-1 = 9)
- For a sample of 100, wanting 99% confidence and just a lower bound, the CC number is: 2.326 (one tail, 1%, n-1 = 99 > 60)

Example:

A chemical process yields, over a sample of runs, products with masses: 2.3mg, 3.9mg, 1.9mg, and 2.7mg.
Calculate the 95% confidence interval for the true mean mass of the yield.

Remember: $\bar{x} \pm CC \times \frac{s}{\sqrt{n}}$

1st step: calculate the mean and the standard deviation: 2.7mg and 0.864mg

2nd step: find the confidence coefficient (two tails, 5%, n-1 = 3):

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail Two Tails	5% 10%	2.5% 5%	1% 2%	0.5% 1%	0.1% 0.2%	0.05% 0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Recall from lectures...

- For example, for an interval of 95% with a sample size of 10, the Confidence Coefficient is: 2.262 (two tails, 5%, n-1 = 9)
- For a sample of 100, wanting 99% confidence and just a lower bound, the CC number is: 2.326 (one tail, 1%, n-1 = 99> 60)

Example:

A chemical process yields, over a sample of runs, products with masses: 2.3mg, 3.9mg, 1.9mg, and 2.7mg.
Calculate the 95% confidence interval for the true mean mass of the yield.

Remember: $\bar{x} \pm CC \times \frac{s}{\sqrt{n}}$

1st step: calculate the mean and the standard deviation: 2.7mg and 0.864mg

2nd step: find the confidence coefficient (two tails, 5%, n-1 =3): 3.182

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail Two Tails	5% 10%	2.5% 5%	1% 2%	0.5% 1%	0.1% 0.2%	0.05% 0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Recall from lectures...

- For example, for an interval of 95% with a sample size of 100, the Confidence Coefficient is: 2.262 (two tails, 5%, n-1 = 9)
- For a sample of 100, wanting 99% confidence and just a lower bound, the CC number is: 2.326 (one tail, 1%, n-1 = 99 > 60)

Example:

A chemical process yields, over a sample of runs, products with masses: 2.3mg, 3.9mg, 1.9mg, and 2.7mg.
Calculate the 95% confidence interval for the true mean mass of the yield.

Remember: $\bar{x} \pm CC \times \frac{s}{\sqrt{n}}$

1st step: calculate the mean and the standard deviation: 2.7mg and 0.864mg

2nd step: find the confidence coefficient (two tails, 5%, n-1 = 3): 3.182

3rd step: do the math: $\bar{x} \pm CC \times \frac{s}{\sqrt{n}} = 2.7\text{mg} \pm 3.182 \frac{0.864\text{mg}}{\sqrt{4}} = 2.7\text{mg} \pm 3.182 \times 0.432\text{mg} = 2.7\text{mg} \pm 1.375\text{mg} = 1.33\text{mg to } 4.07\text{mg}$

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail Two Tails	5% 10%	2.5% 5%	1% 2%	0.5% 1%	0.1% 0.2%	0.05% 0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Recall from lectures...

- For example, for an interval of 95% with a sample size of 100, the Confidence Coefficient is: 2.262 (two tails, 5%, n-1 = 9)
- For a sample of 100, wanting 99% confidence and just a lower bound, the CC number is: 2.326 (one tail, 1%, n-1 = 99 > 60)

Example:

A chemical process yields, over a sample of runs, products with masses: 2.3mg, 3.9mg, 1.9mg, and 2.7mg.
Calculate the 95% confidence interval for the true mean mass of the yield.

Remember: $\bar{x} \pm CC \times \frac{\sigma}{\sqrt{n}}$

1st step: calculate the mean and the **standard deviation**: 2.7mg and 0.864mg

2nd step: find the confidence coefficient (two tails, 5%, n-1 = 3): 3.182

3rd step: do the math: $\bar{x} \pm CC \times \frac{s}{\sqrt{n}} = 2.7\text{mg} \pm 3.182 \frac{0.864\text{mg}}{\sqrt{4}} = 2.7\text{mg} \pm 3.182 \times 0.432\text{mg} = 2.7\text{mg} \pm 1.375\text{mg} = 1.33\text{mg to } 4.07\text{mg}$

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail Two Tails	5% 10%	2.5% 5%	1% 2%	0.5% 1%	0.1% 0.2%	0.05% 0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Question 1

The purpose of inferential statistics is to:

1. to summarize observations
2. to describe sets of data
3. use information from a sample to infer results about a population
4. to measure everything in a population

Question 1

The purpose of inferential statistics is to:

1. to summarize observations
2. to describe sets of data
3. **use information from a sample to infer results about a population**
4. to measure everything in a population

Question 2

Which is a primary data source?

1. a report by the Council detailing the number of car parking spaces around York
2. the results of a survey you've carried out into car ownership among your friends
3. the national census
4. the tables of data for questions in formative quizzes

Question 2

Which is a primary data source?

1. a report by the Council detailing the number of car parking spaces around York
2. **the results of a survey you've carried out into car ownership among your friends**
3. the national census
4. the tables of data for questions in formative quizzes

Remember! **Primary** data is any **you collect yourself** for this purpose. **Secondary** data is any data **someone else** has already collected.

Question 3

Which of the following is a description of the **Random sampling method**?

- Every item in the population has exactly the same chance to be sampled, and no other constraints apply.
- Every kth item in the population is chosen for the sample. (k can be 3, 5, 10 etc.)
- The population is subdivided into groups that are as alike as possible, often through multiple subdivisions. Items from some of these subdivisions form the sample.
- Items for the sample are picked from groups which are as unlike as possible.

Question 3

Which of the following is a description of the **Random sampling method**?

- **Every item in the population has exactly the same chance to be sampled, and no other constraints apply.**
- Every kth item in the population is chosen for the sample. (k can be 3, 5, 10 etc.)
- The population is subdivided into groups that are as alike as possible, often through multiple subdivisions. Items from some of these subdivisions form the sample.
- Items for the sample are picked from groups which are as unlike as possible.

Question 4

Open-ended leading questions in a questionnaire are:

- Looking for a specific answer.
- Difficult to process.
- Liable to introduce bias.
- All of these.

Question 4

Open-ended leading questions in a questionnaire are:

- Looking for a specific answer.
- Difficult to process.
- Liable to introduce bias.
- **All of these.**

Question 5

Which of the following is NOT a brilliant thing about the Central Limit Theorem?

- Samples means are drawn from the same distribution as the population.
- The standard error of the data means bigger samples tend to be better.
- The mean of the sample means is the same as the population mean.
- Sufficiently large samples have means drawn from a normal distribution, regardless of the population distribution.

Question 5

Which of the following is NOT a brilliant thing about the Central Limit Theorem?

- **Samples means are drawn from the same distribution as the population.**
- The standard error of the data means bigger samples tend to be better.
- The mean of the sample means is the same as the population mean.
- Sufficiently large samples have means drawn from a normal distribution, regardless of the population distribution.

Question 5

Which of the following is NOT a brilliant thing about the Central Limit Theorem?

- **Samples means are drawn from the same distribution as the population.**
- The standard error of the data means bigger samples tend to be better.
- The mean of the sample means is the same as the population mean.
- Sufficiently large samples have means drawn from a normal distribution, regardless of the population distribution.

Hint! It is crucial to study the theory very well to answer the formative and the summative questions!

- **Primary vs Secondary data,**
- **Types of sampling:** Random, Systematic, Stratified, Cluster,
- **Data collection:** closed-ended vs open-ended questions

Question 6

A sample of 88 items returns a mean of 56.5 and a standard deviation of 11.9. What's the 99% confidence **upper bound** for the population mean?

Remember: $\bar{x} + CC \times \frac{s}{\sqrt{n}}$

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Question 6

A sample of 88 items returns a mean of 56.5 and a standard deviation of 11.9. What's the 99% confidence upper bound for the population mean?

Remember: $\bar{x} + CC \times \frac{s}{\sqrt{n}}$

1st step: we know the mean and the standard deviation: 56.5 and 11.9

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Question 6

A sample of 88 items returns a mean of 56.5 and a standard deviation of 11.9. What's the 99% confidence upper bound for the population mean?

Remember: $\bar{x} + CC \times \frac{s}{\sqrt{n}}$

1st step: we know the mean and the standard deviation: 56.5 and 11.9

2nd step: find the confidence coefficient (one tail, 1%, n-1 =87):

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Question 6

A sample of 88 items returns a mean of 56.5 and a standard deviation of 11.9. What's the 99% confidence **upper bound** for the population mean?

Remember: $\bar{x} + CC \times \frac{s}{\sqrt{n}}$

1st step: we know the mean and the standard deviation: 56.5 and 11.9

2nd step: find the confidence coefficient (**one tail**, 1%, n-1 =87):

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Question 6

A sample of 88 items returns a mean of 56.5 and a standard deviation of 11.9. What's the 99% confidence upper bound for the population mean?

Remember: $\bar{x} + CC \times \frac{s}{\sqrt{n}}$

1st step: we know the mean and the standard deviation: 56.5 and 11.9

2nd step: find the confidence coefficient (one tail, 1%, n-1 =87): 2.326

3rd step: do the math:

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Question 6

A sample of 88 items returns a mean of 56.5 and a standard deviation of 11.9. What's the 99% confidence upper bound for the population mean?

Remember: $\bar{x} + CC \times \frac{s}{\sqrt{n}}$

1st step: we know the mean and the standard deviation: 56.5 and 11.9

2nd step: find the confidence coefficient (one tail, 1%, n-1 =87): 2.326

3rd step: do the math: $56.5 + 2.326 \times (11.9 / \sqrt{88}) = 59.45$

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Question 7

A sample is 98.1, 78.9, 61, 67.8, and 12.4. Calculate the 99% confidence **lower bound** of the population mean.

Remember: $\bar{x} - CC \times \frac{s}{\sqrt{n}}$

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Question 7

A sample is 98.1, 78.9, 61, 67.8, and 12.4. Calculate the 99% confidence **lower bound** of the population mean.

Remember: $\bar{x} - CC \times \frac{s}{\sqrt{n}}$

1st step: calculate the mean and the standard deviation:

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Question 7

A sample is 98.1, 78.9, 61, 67.8, and 12.4. Calculate the 99% confidence lower bound of the population mean.

Remember: $\bar{x} - CC \times \frac{s}{\sqrt{n}}$

1st step: calculate the mean and the standard deviation: 63.64 and 31.9

*(We need to use s, not σ)

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Question 7

A sample is 98.1, 78.9, 61, 67.8, and 12.4. Calculate the 99% confidence lower bound of the population mean.

Remember: $\bar{x} - CC \times \frac{s}{\sqrt{n}}$

1st step: calculate the mean and the standard deviation: 63.64 and 31.9

*(We need to use s, not σ)

2nd step: find the confidence coefficient (one tail, 1%, n-1 =4):

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Question 7

A sample is 98.1, 78.9, 61, 67.8, and 12.4. Calculate the 99% confidence lower bound of the population mean.

Remember: $\bar{x} - CC \times \frac{s}{\sqrt{n}}$

1st step: calculate the mean and the standard deviation: 63.64 and 31.9

*(We need to use s, not σ)

2nd step: find the confidence coefficient (one tail, 1%, n-1 =4): 3.747

3rd step: do the math:

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Question 7

A sample is 98.1, 78.9, 61, 67.8, and 12.4. Calculate the 99% confidence lower bound of the population mean.

Remember: $\bar{x} - CC \times \frac{s}{\sqrt{n}}$

1st step: calculate the mean and the standard deviation: 63.64 and 31.9

*(We need to use s, not σ)

2nd step: find the confidence coefficient (one tail, 1%, n-1 =4): 3.747

3rd step: do the math: $63.64 - 3.747 \times (31.9 / \sqrt{5}) = 10.19$

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Question 8

It is found in a sample of 118 items that 9 of them don't work properly...What's the 95% confidence upper bound for the population proportion ?

Hint! We are in a Bernoulli world! Instead of $\bar{x} + CC \times \frac{\sigma}{\sqrt{n}}$, the interval we are looking for

is: $p + CC \times \sqrt{\frac{p(1-p)}{n}}$

1st step: calculate the mean and the variance:

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Question 8

It is found in a sample of 118 items that 9 of them don't work properly...What's the 95% confidence upper bound for the population proportion ?

Hint! We are in a Bernoulli world! Instead of $\bar{x} + CC \times \frac{\sigma}{\sqrt{n}}$, the interval we are looking for

is: $p + CC \times \sqrt{\frac{p(1-p)}{n}}$

1st step: calculate the mean and the variance: 0.076 and 0.070

2nd step: find the confidence coefficient (one tail, 5%, n-1 =117):

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Question 8

It is found in a sample of 118 items that 9 of them don't work properly...What's the 95% confidence upper bound for the population proportion ?

Hint! We are in a Bernoulli world! Instead of $\bar{x} + CC \times \frac{\sigma}{\sqrt{n}}$, the interval we are looking for

is: $p + CC \times \sqrt{\frac{p(1-p)}{n}}$

1st step: calculate the mean and the variance: 0.076 and 0.070

2nd step: find the confidence coefficient (one tail, 5%, n-1 =117): 1.645

3rd step: do the math:

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Question 8

It is found in a sample of 118 items that 9 of them don't work properly...What's the 95% confidence upper bound for the population proportion ?

Hint! We are in a Bernoulli world! Instead of $\bar{x} + CC \times \frac{\sigma}{\sqrt{n}}$, the interval we are looking for

is: $p + CC \times \sqrt{\frac{p(1-p)}{n}}$

1st step: calculate the mean and the variance: 0.076 and 0.070

2nd step: find the confidence coefficient (one tail, 5%, n-1 =117): 1.645

3rd step: do the math: $0.076 + 1.645 \times \sqrt{0.070/118} = 0.116$

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Question 9

A researcher wishes to have 21% accuracy and 95% confidence when estimating a proportion, which they have previously estimated to be 31%. How big should their sample be?

Hint! We are in a Bernoulli world! $p \pm CC \times \sqrt{\frac{p(1-p)}{n}}$

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Question 9

A researcher wishes to have 21% accuracy and 95% confidence when estimating a proportion, which they have previously estimated to be 31%. How big should their sample be?

Hint! We are in a Bernoulli world! $p \pm CC \times \sqrt{\frac{p(1-p)}{n}}$

1st step: re-arrange the formula: $n = \frac{CC^2 p(1-p)}{a^2}$ where $a = CC \times \sqrt{\frac{p(1-p)}{n}} = 0.21$

2nd step: find the confidence coefficient (two tail, 5%, n>60+):

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Question 9

A researcher wishes to have 21% accuracy and 95% confidence when estimating a proportion, which they have previously estimated to be 31%. How big should their sample be?

Hint! We are in a Bernoulli world! $p \pm CC \times \sqrt{\frac{p(1-p)}{n}}$

1st step: re-arrange the formula: $n = \frac{CC^2 p(1-p)}{a^2}$ where $a = CC \times \sqrt{\frac{p(1-p)}{n}} = 0.21$

2nd step: find the confidence coefficient (two tail, 5%, n>60+): 1.96

3rd step: do the math:

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Question 9

A researcher wishes to have 21% accuracy and 95% confidence when estimating a proportion, which they have previously estimated to be 31%. How big should their sample be?

Hint! We are in a Bernoulli world! $p \pm CC \times \sqrt{\frac{p(1-p)}{n}}$

1st step: re-arrange the formula: $n = \frac{CC^2 p(1-p)}{a^2}$ where $a = CC \times \sqrt{\frac{p(1-p)}{n}} = 0.21$

2nd step: find the confidence coefficient (two tail, 5%, n>60+): 1.96

3rd step: do the math: $n = \frac{1.960^2 0.31(1-0.31)}{0.21^2} = 18.63 = 19$

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Question 10

A researcher wishes to have an accuracy of 7.1cm with a confidence of 95%. They have previously estimated s to be 10cm. How big should their sample be?

Hint! We return to a continuous distribution world! $\bar{x} \pm CC \times \frac{s}{\sqrt{n}}$

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Question 10

A researcher wishes to have an accuracy of 7.1cm with a confidence of 95%. They have previously estimated s to be 10cm. How big should their sample be?

Hint! We return to a continuous distribution world! $\bar{x} \pm CC \times \frac{s}{\sqrt{n}}$

1st step: re-arrange the formula: $a = CC \times SE = \frac{CCs}{\sqrt{n}}$, so $n = \left(\frac{CCs}{a}\right)^2$

2nd step: find the confidence coefficient (two tail, 5%, n>60+):

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Question 10

A researcher wishes to have an accuracy of 7.1cm with a confidence of 95%. They have previously estimated s to be 10cm. How big should their sample be?

Hint! We return to a continuous distribution world! $\bar{x} \pm CC \times \frac{s}{\sqrt{n}}$

1st step: re-arrange the formula: $a = CC \times SE = \frac{CCs}{\sqrt{n}}$, so $n = \left(\frac{CCs}{a}\right)^2$

2nd step: find the confidence coefficient (two tail, 5%, n>60+): 1.96

3rd step: do the math:

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Question 10

A researcher wishes to have an accuracy of 7.1cm with a confidence of 95%. They have previously estimated s to be 10cm

Hint! We return to a continuous distribution world! $\bar{x} \pm CC \times \frac{s}{\sqrt{n}}$

1st step: re-arrange the formula: $a = CC \times SE = \frac{CCs}{\sqrt{n}}$, so $n = \left(\frac{CCs}{a}\right)^2$

2nd step: find the confidence coefficient (two tail, 5%, n>60+): 1.96

3rd step: do the math: $n = \left(\frac{1.96 \times 10}{7.1}\right)^2 = 7.6 = 8$

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Sum up

1. **Inference**: using a **sample** to infer results about a **population**
2. Methods and Types of sampling
3. The Central Limit Theorem
4. t-distribution
5. Construct a confidence interval
6. Upper and lower bounds